

Create Azure Local VMs Enabled by Azure Arc



Agenda

- **Objective:** Review deploying Azure Local VMs Enabled by Azure Arc
- **Scope:** Architecture, VM Types, Deployment Flow, and Management Capabilities



VM Types on Azure Local

- **Azure Local supports three types of VMs:**
 - Azure Local VMs enabled by Azure Arc
 - Azure Arc-enabled Servers
 - Unmanaged VMs

VM Type Comparison

- **Azure Local VMs enabled by Azure Arc**
 - Created via **Azure portal**, **Azure CLI**, or **ARM/Bicep**
 - Fully projected into Azure via the **Azure Arc resource bridge**
 - Support lifecycle operations: **start**, **stop**, **resize**, **disk attach**, **NIC attach**, etc.
- **Azure Arc-enabled Servers**
 - Physical or virtual machines connected to Azure using the **Connected Machine Agent**
 - No platform-level VM lifecycle controls
 - Support management & governance extensions only
- **Unmanaged VMs**
 - Created solely with **on-prem management tools**
 - Not synchronized with Azure
 - No Azure-based management

VM Provisioning Comparison

- **Arc-enabled VMs:** Provisioned in Azure; deployed locally
- **Arc-enabled servers:** Created via Hyper-V/Failover Cluster Manager/System Center and connected to Azure
- **Unmanaged VMs:** Created via Hyper-V/Failover Cluster Manager/System Center

VM Management Capability Comparison

- **Arc-enabled Azure Local VMs support:**
 - Start / Stop / Restart / Pause
 - Configuration changes (CPU, Memory)
 - Disk operations (add/remove/expand)
 - NIC operations (add/remove)
 - GPU attach/detach
 - Azure Monitoring, Defender for Cloud, Policies, Extensions
- **Other VM types support partial or no lifecycle management.**

Core Component: Azure Arc Resource Bridge

- Prepackaged VM deployed automatically during Azure Local installation
- Projects local resources into Azure
- Acts as the management cluster
- Hosts Arc-enabled Kubernetes instance and the Kubernetes VM extension
- Must NOT be deleted unless decommissioning the cluster

Core Component: Kubernetes VM Extension

- Implements VM lifecycle actions
- **Handles:**
 - Power operations
 - Disk attach/detach
 - NIC attach/detach
 - GPU operations
- **Executes ARM-triggered operations on the local cluster**

Core Component: Custom Location

- Represents your Azure Local instance in Azure
- Used as the deployment target for VMs, NICs, disks, images
- One-to-one mapping with the resource bridge namespace

Core Component: Infrastructure Logical Network

- Created automatically during deployment
- Used by the Arc resource bridge
- Not user-modifiable

Azure Local VM Management Workflow

1. Deploy Azure Local → resource bridge + custom location created
2. Assign RBAC roles
 - Azure Stack HCI Administrator
 - Azure Stack HCI VM Contributor
 - Azure Stack HCI VM Reader
3. Create supporting VM resources:
 - Storage paths
 - VM images
 - Logical networks
 - Network interfaces
4. Create a VM
5. Use Azure Portal/CLI for all supported operations

Creating Storage Paths

- **Storage paths store:**
 - VM configuration files
 - OS & data disks
 - Image files
- **Must reside on Cluster Shared Volumes (CSV)**
- **Should not use Infrastructure volumes**

VM Images: Supported Sources

You can create VM images from:

- Azure Marketplace
- Azure Compute Gallery
- Azure Storage Account
- Local share (on-prem)
- Existing Azure Local VM's OS disk

VM Image Requirements

- Only English (en-us) VHDs supported
- Must be generalized
- For VHDX:
 - Must be Gen 2
 - Must have Secure Boot enabled

Network Interfaces Overview

- Must be associated with a logical network
- NIC can be:
 - Static (automatic from an IP pool or manual assignment)
 - DHCP
- When created during VM deployment, NICs inherit configuration from the logical network

VM Creation Workflow

1. Click + Create VM
2. Enter:
 - VM name
 - Custom location
 - Security type
 - Storage path
3. Select image
4. Configure:
 - vCPU
 - Memory
 - Extensions
 - Networking
5. Configure proxy (optional)
6. Add disks (optional)
 - Disk name
 - Capacity
 - Provisioning type (static/dynamic)
 - Storage path
7. Add NICs (optional)
8. Review + create

Proxy for Azure Local VMs

- Public endpoint
- Proxy server
 - HTTP proxy
 - HTTPS proxy
 - No-proxy exclusions
 - Certificate file for trust establishment

VM Extensions (Optional)

Arc-enabled VMs support:

- Guest management
- Azure Monitor
- Defender for Cloud
- Custom extensions

Domain Join (Windows Only)

- Supports Active Directory domain join
- Requires:
 - UPN of domain user
 - Password
 - Domain or OU specification

Supported Cloud Initiated Azure Local VMs Operations

- Performed from the Azure Portal or Azure CLI:
 - Create
 - Resize CPU/memory
 - Add/remove disks
 - Add/remove NICs
 - Attach/detach GPU
 - Start/stop/restart/pause
 - Enable guest management
 - Install extensions
 - Connect via SSH/RDP
- Performing these operations via local tools risks state conflicts

Supported Local-Only Azure Local VMs Operations

- **Performed via Hyper-V Manager/Failover Cluster Manager/PowerShell:**
 - Rename VM
 - MAC address changes
 - NUMA config
 - Boot order changes
 - Secure boot enable
 - Checkpoints (HCI 2504+)
 - SCSI controller modifications
- **These changes do not sync to Azure**

Unsupported Operations

- **Avoid entirely:**
 - Changing VLAN ID
 - Storage live migration
 - Cloning or copying VM
 - Changing disk type
 - Cross-cluster live migration
- **These operations break VM management state in Azure and can render VMs unmanageable.**

Managed Identities

- Arc-enabled VMs include a system-assigned managed identity
- Allow access to:
 - Azure Key Vault
 - Azure Storage
 - Other identity-protected endpoints

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